

12 Approximating Expected Values

- The expected value of a random variable X is its *probability-weighted average value* $E(X)$
- $E(X)$ can be interpreted as the *long run average value* of X
- We can approximate $E(X)$ by simulating many values of the random variable and computing the average (mean) in the usual way: add up the simulated values of X and divide by the number of simulated values.

Example 12.1. Let X be the number of vehicles crashes involving fatalities that occur in San Luis Obispo County in a randomly selected week. Suppose that X has distribution

x	$P(X = x)$
0	0.589
1	0.312
2	0.083
3	0.014
4	0.002

1. Interpret the probabilities as long run relative frequencies in this context
2. Sketch a spinner for simulating values of X .
3. Describe in detail how you could use simulation to approximate $E(X)$.

4. Compute $E(X)$ and compare to the simulated approximation.

5. Interpret $E(X)$ in context.

6. Suppose you want to approximate $E(X^2)$. (We'll see why soon.) Donny Don't says: "Just square the value from part 3". Is that true? If not, how would you use simulation to approximate $E(X^2)$?

7. Compute $E(X^2)$. Is it equal to $E(X)^2$?

- In general the order of transforming and averaging is not interchangeable.
- Whether in the short run or the long run, in general

$$\begin{aligned}\text{Average of } g(X) &\neq g(\text{Average of } X) \\ \text{Average of } g(X, Y) &\neq g(\text{Average of } X, \text{Average of } Y)\end{aligned}$$

- In terms of expected values (long run averages), in general

$$\begin{aligned}E(g(X)) &\neq g(E(X)) \\ E(g(X, Y)) &\neq g(E(X), E(Y))\end{aligned}$$

Example 12.2. Continuing Example 12.1. Suppose that the number of fatal crashes in any one week period follows the distribution in Example 12.1, independently from week to to week. Now consider a two week period, say this week and next week. Let X be the number of fatal crashes in the first week and Y in the second week, so that $X + Y$ is the total number of fatal crashes in the two week period.

1. Are X and Y the same random variable?
2. Do X and Y have the same distribution? Explain.
3. What is $E(Y)$. Why?
4. Describe how you would conduct a simulation to approximate the distribution of $X + Y$.
5. Describe how you would use simulation to approximate $E(X + Y)$.
6. Use simulation to approximate $E(X)$, $E(Y)$, and $E(X + Y)$. How are they related?
7. Compute $P(X + Y = 0)$.

8. Compute $P(X + Y = 1)$.

9. The following table represents the joint distribution of X and Y . Explain how this table was constructed.

x, y	0	1	2	3	4
0	0.3469	0.1838	0.0489	0.0082	0.0012
1	0.1838	0.0973	0.0259	0.0044	0.0006
2	0.0489	0.0259	0.0069	0.0012	0.0002
3	0.0082	0.0044	0.0012	0.0002	0.0000
4	0.0012	0.0006	0.0002	0.0000	0.0000

10. Compute the distribution of $X + Y$.

11. Compute $E(X + Y)$. Interpret the value in context.

12. How is $E(X + Y)$ related to $E(X)$ and $E(Y)$?

- In general the order of transforming and averaging is not interchangeable.
- However, the order is interchangeable for *linear* transformations.
- **Linearity of averages.:** If X and Y are random variables and a and b are non-random constants, whether in the short run or the long run, $\begin{align} \text{Average of } a + bX \text{ } \mathcal{E} \\ = a + b(\text{Average of } X) \end{align}$ $\text{Average of } X + Y \text{ } \mathcal{E} = \text{Average of } X + \text{Average of } Y$ $\text{End}\{align\}$
- **Linearity of expected value.:** If X and Y are random variables and m and b are non-random constants,

$$E(mX + b) = mE(X) + b$$

$$E(X + Y) = E(X) + E(Y)$$

- The expected value of the sum of X and Y is the sum of the expected value of X and the expected value of Y *regardless of the relationship between X and Y .*

Example 12.3. Continuing Example 12.2.

1. Explain how we could use simulation to approximate the conditional distribution of $X + Y$ given $X = 1$.
2. Explain how we could use simulation to approximate $E(X + Y | X = 1)$.
3. Compute the conditional distribution of $X + Y$ given $X = 1$.
4. Compute $E(X + Y | X = 1)$